SUPPLEMENTARY MATERIAL

Supplementary material: Certified Action Sets and Decision Limits in Surrogate-Assisted Engineering Optimization

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Working manuscript. Affiliations b and c remain to be supplied.

S0. Evidence Map and Reproducibility Checklist

The supplement is organized by the claims made in the main article. Scientific results, resource measurements, and recovery records are separated so that interrupted physical calls and implementation checks cannot be mistaken for optimization evidence.

S0.1. *Evidence map*

1. **Calibration and decision utility:** S1-S2, Tables S1-S7, and Figures S1-S4 provide declaration counts, failure fractions, common-history diagnostics, and common-state utility.
2. **External descriptive context:** S3, Tables S8-S12, and Figures S5-S6 report full BBOB endpoints and exclusive process costs under distinct search histories.
3. **Transfer evidence:** S4-S5, Tables S13-S19, and Figures S7-S10 cover LassoBench and the 300-dimensional experiment.
4. **Sensitivity:** S6-S7, Tables S20-S24, and Figures S11-S13 report model, training-cap, and total-budget sensitivity.
5. **Information order and accounting:** S7.3-S8.3 and Table S25 preserve finite-element replay, source hashes, recovery ledgers, and matched-initialization evidence.
6. **Additional mechanism and engineering evidence:** S9-S10, Tables S26-S32 and Figure S14 contain the frozen GP intervention, delayed policy outcomes, force-position transfer, fixed-policy comparisons and full revision cost boundary.

S0.2. *Reproducibility checklist*

1. **Runtime:** Windows 11, Python 3.12.13, and dependency lock files for the main, application, GP, and reporting environments.
2. **Hardware:** AMD Ryzen 7 9700X, 16 logical threads, and approximately 31.1 GiB physical memory; numerical-library threads are limited to one per worker.
3. **Randomness:** Purpose-specific seed maps cover initialization, candidate generation, mode selection, optimization repeats, and bootstrap analyses.

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4. **Protocols and parameters:** Frozen JSON configurations retain budgets, function and instance lists, model settings, and entry-point arguments.
5. **Information order:** PRE/LABEL journals, prediction-time training indices, candidate arrays, and separate diagnostic-label archives are retained.
6. **Integrity:** Source fingerprints, dependency hashes, raw-file hashes, endpoint replay, and expected aggregate tables are archived.
7. **Data and applications:** BBOB/COCO identifiers, fixed LassoBench datasets and splits, and the deterministic EngiBench finite-element parameterization are recorded.
8. **Recovery:** Incomplete units, canonical-versus-physical call ledgers, rerun seeds, and exclusion rules are archived separately.

The source repository is private, so its URL alone does not provide reviewer access. The scoped data-and-code attachment contains the complete inputs and certificates for the current GP action-set verification, plus selected frozen decision records. Complete raw reconstruction of the supporting paired-grid check is currently unavailable: 22 of 25 audit-listed files have been recovered, while the final recovery journal, response collection and status remain missing. The historical outputs are retained with this limitation; they cannot replace the absent raw records. The current preservation status is detailed in Section S13. The authors report a local copy of approximately 13.6 GB of historical outputs, but it has not been fully inspected in this preparation and is not part of the scoped attachment. Full historical availability is not guaranteed. Third-party implementations and publisher templates retain their own terms. The detailed historical audit trail remains in S8; this map is a navigation layer rather than a substitute for those records.

S1. Development diagnostics and common-state evidence

S1.1. *Exploration, failure, and sparse declarations*

PEM-T explores approximately 36.6% and 33.4% of post-initialization evaluations, compared with approximately 10% for the fixed control. Its observed failure fractions are approximately 9.9% and 9.5%. Being near the nominal monitoring level does not correspond to better endpoints.

Table S1. Joint closed-loop decision indicators, 72 runs per dimension/method

D	Method	Exploration	Failure	Claims	False	Pooled FDP
10	PEM_ST	0.3574	0.1062	7	4	0.5714
10	PEM_T	0.3664	0.0986	3	1	0.3333
10	error_only	0.1574	0.0847	4	2	0.5000
10	fixed01	0.0995	0.0843	4	2	0.5000
10	stagnation_only	0.2338	0.0972	2	0	0.0000
20	PEM_ST	0.3478	0.1098	32	2	0.0625
20	PEM_T	0.3335	0.0946	0	0	NA
20	error_only	0.1475	0.0864	8	4	0.5000
20	fixed01	0.1017	0.0868	10	9	0.9000
20	stagnation_only	0.2394	0.0951	1	1	1.0000

PEM-T makes only three declarations in dimension 10, one false, and no declarations

in dimension 20. The latter FDP is NA, not zero. PEM-ST makes 32 declarations with two false in dimension 20, but its endpoint remains worse. Table S1's pooled declaration ratio is distinct from a mean over runs with nonzero declarations. These sparse denominators do not establish stable conditional success probabilities.

The numbers of realized improvements and false claims within each declared subset are shown in Figure S1.

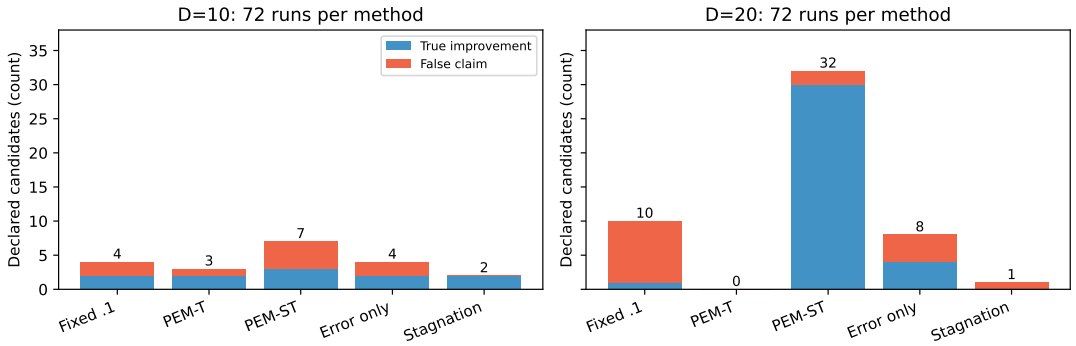


Figure S1. Counts of actual improvement and false claims on declared candidates. A zero column denotes no declarations, not an established reliability advantage.

Figure S1 exposes both the rarity and composition of improvement declarations. PEM-T makes only three declarations in dimension 10 and none in dimension 20, whereas PEM-ST and the fixed controls produce larger declared subsets. Blue and orange segments distinguish realized improvements from false claims. A missing bar therefore represents an empty conditional sample, while a tall orange segment identifies a declaration rule whose apparent candidates often fail to improve the incumbent.

Failure-only feedback is less detrimental than full feedback on the endpoint summary but still does not achieve an overall benefit over fixed .1. Its combined removal of width and stagnation cannot isolate those terms' independent contributions.

S1.2. *Common-history diagnostics*

Table S2 uses the same T bound within each frozen history/model/pool. Under fixed .3 histories, failures at the actually selected point exceed pool failures. Under greedy histories, their ordering reverses. The observation supports history-dependent differences among monitoring objects and does not support a universal direction of selection-induced undercoverage. Each dimension/history has 288 frozen states; each state contributes one actually selected candidate, 256 pool candidates, and 26 top-10% candidates.

Table S2. Temporal bounds across monitoring objects

D	History	Pool	Top 10%	Actually selected
10	fixed03	0.0425	0.0290	0.1250
10	greedy	0.1070	0.0395	0.0868
10	pool_random	0.0773	0.0944	0.0521
20	fixed03	0.0481	0.0423	0.1424
20	greedy	0.1691	0.0843	0.0938
20	pool_random	0.0935	0.1062	0.1042

Failure fractions for the selected candidate, full pool, and predicted top 10% are

compared in Figure S2.

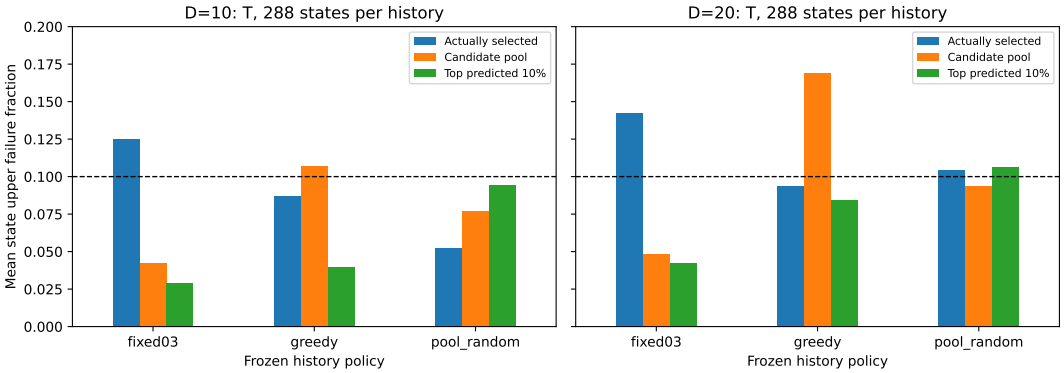


Figure S2. State-averaged one-sided failure fractions under frozen common histories. The dashed target is .1. Pool labels are correlated and are not treated as independent problems.

Figure S2 compares three monitored objects at identical frozen states: the actually selected candidate, the entire candidate pool, and the top 10% by predicted value. Under fixed 03 histories, the selected-candidate bar is highest; under greedy histories, the pool and top-10% bars become higher. The dashed 0.1 line is crossed in different directions across histories, which supports reporting the monitored object and its generating policy together.

Table S3 compares calibrators on the actually selected points, with the three histories equally weighted. Spatial localization lowers some diagnostic losses and margins; this is a common-history estimation result, not a closed-loop optimization gain or a coverage guarantee. Approximately one quarter of the dimension-10 checkpoint states fall back because of support, whereas these dimension-20 checkpoints do not.

Table S3. Calibrators on actually selected points

D	Calibrator	Quantile loss	Margin	ESS	Fallback
10	G	0.1355	1.1044	45.5000	0.0000
10	S	0.1241	0.9463	38.8140	0.2500
10	ST	0.1187	0.9175	32.1634	0.2500
10	T	0.1284	1.0262	37.7288	0.2500
20	G	0.1465	0.9365	74.7500	0.0000
20	S	0.1322	0.7874	60.5382	0.0000
20	ST	0.1307	0.7643	42.6863	0.0000
20	T	0.1409	0.9114	52.3172	0.0000

Pool-declaration subsets are also sparse: T has only 13 declared candidates under dimension-10 fixed .3 histories, and one under dimension-20 fixed .3 histories. Full state, nonempty-state, candidate, and failure denominators are publicly retained in the diagnostic CSV files. Stable reliability is not inferred from a small or empty subset.

S1.3. *Common-state immediate utility*

Table S4 reports normalized true improvement at one chosen candidate from the same frozen pool. T upper-bound selection is identical to mean greedy and is not repeated

as an independent column. Full feedback has lower mean immediate improvement than mean greedy and than the predeclared fixed .3 rule in all six dimension/history cells.

Table S4. Common-state normalized one-step improvement

D	History	Mean	Fixed .3	Feedback
10	fixed03	0.0260	0.0219	0.0213
10	greedy	0.0310	0.0257	0.0251
10	pool_random	0.1240	0.0925	0.0867
20	fixed03	0.0350	0.0238	0.0172
20	greedy	0.0541	0.0433	0.0381
20	pool_random	0.1326	0.0871	0.0795

Immediate utility does not exclude a delayed benefit from exploration; however, the closed-loop development experiment supplies no evidence of that benefit for this prototype. It is therefore unsupported to state that additional distance exploration repaired the model.

S2. Independent validation diagnostics and external references

S2.1. *History, declarations, and effective support*

The history-interaction estimates are D=10: +0.0851 (Holm p=0.00149); D=20: +0.1225 (Holm p=0.00065); D=40: +0.1449 (Holm p=0.00170). This is the fixed .3 selected-minus-pool difference minus the greedy corresponding difference. Across policies, identical seeds do not imply identical candidate pools. These differences describe the coupled history/policy system rather than a fixed-pool causal effect.

Table S5. Temporal pool and actually selected failure fractions, under the prespecified state/run/instance/function aggregation.

D	History	Pool failure	Selected failure
10	fixed03	0.0479	0.1215
10	greedy	0.1018	0.0903
10	pool_random	0.0859	0.0660
20	fixed03	0.0442	0.0833
20	greedy	0.1494	0.0660
20	pool_random	0.0900	0.0729
40	fixed03	0.0505	0.0799
40	greedy	0.1850	0.0694
40	pool_random	0.0978	0.0694

The corresponding selected-minus-pool contrasts are shown in Figure S3.

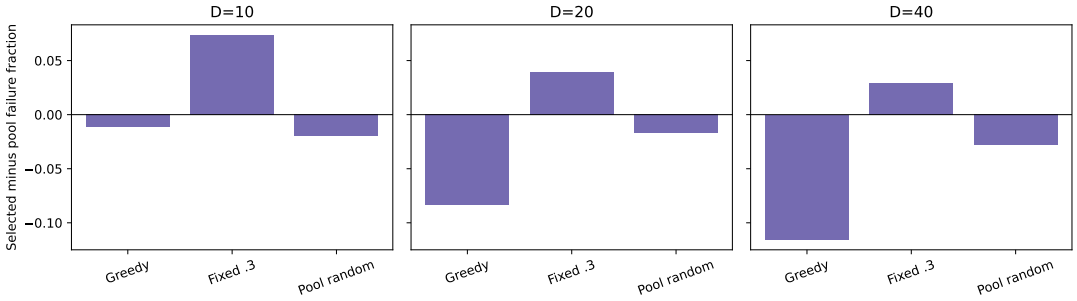


Figure S3. Selected-minus-pool failures by preselected history. A reversal is retained rather than interpreted as universal selection deterioration.

Figure S3 expresses the same comparison as a signed difference (selected minus pool). Fixed03 bars are positive and greedy bars are negative in every dimension; random-pool bars stay nearer zero. The repeated sign reversal demonstrates that selected-point coverage cannot be summarized independently of the history-generating policy.

Table S6. Exploration, failures, and declaration denominators in independent closed-loop runs. FDP is pooled false/claims; no-claim FDP is NA.

D	Method	Exploration	Failure	Claims	False	Pooled FDP
10	PEM_ST	0.3936	0.1052	48	23	0.4792
10	PEM_T	0.3967	0.1008	31	20	0.6452
10	fixed01	0.0949	0.0812	35	21	0.6000
20	PEM_ST	0.3687	0.1078	158	36	0.2278
20	PEM_T	0.3644	0.0942	34	27	0.7941
20	fixed01	0.1018	0.0833	48	32	0.6667
40	PEM_ST	0.3701	0.1196	870	98	0.1126
40	PEM_T	0.3435	0.0964	155	30	0.1935
40	fixed01	0.1007	0.0930	388	71	0.1830

For PEM-T, D=10: 31 declarations, 20 false, pooled FDP 64.5%; D=20: 34 declarations, 27 false, pooled FDP 79.4%; D=40: 155 declarations, 30 false, pooled FDP 19.4%. An approximately .1 overall failure fraction does not imply an approximately .1 false-declaration fraction. These small declaration denominators do not establish stable per-declaration probabilities. ST lowers the descriptive selected-point quantile loss in all nine dimension/history cells, yet its closed-loop endpoint remains worse than fixed .1 in every dimension. These are separate diagnostic and trajectory comparisons, not an identified causal relationship.

Table S7. Descriptive state-mean support across the three preselected history policies and four checkpoints.

D	Calibrator	State mean ESS	State mean fallback
10	ST	32.9284	0.2500
10	T	37.7288	0.2500
20	ST	43.6167	0.0000
20	T	52.3172	0.0000
40	ST	49.4855	0.0004
40	T	58.5602	0.0000

Temporal and spatiotemporal effective support across checkpoint order is shown in Figure S4.

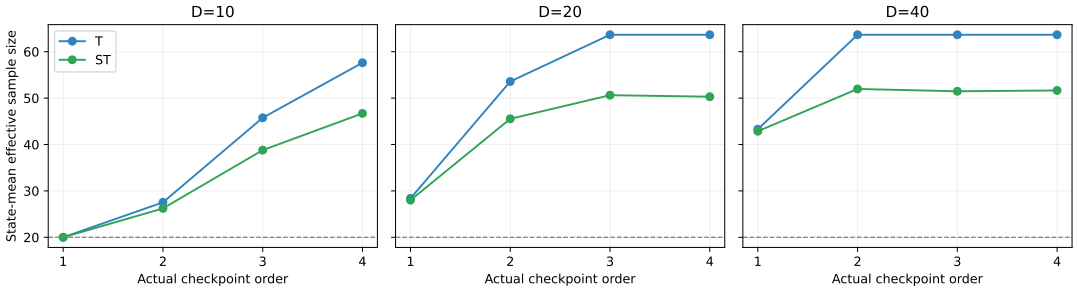


Figure S4. T/ST effective support at the four actual checkpoint orders. The threshold is an engineering guard and does not guarantee coverage.

Figure S4 shows that effective support grows with checkpoint order for both temporal (T) and spatiotemporal (ST) weighting. T retains more effective observations than ST after localization, and the gap is largest in dimensions 20 and 40. Both curves remain above the support guard after the earliest checkpoint, but the plateaus emphasize that passing the guard is a numerical eligibility condition rather than evidence of calibrated coverage.

The first diagnostic checkpoints are 41/100, 61/200, and 101/400. Sparse declared-pool subsets disclose nonempty states, candidate counts, and failures separately. Localized quantile-loss, margin, ESS, and fallback differences and the five fixed BBOB structure groups are available in full secondary tables; no favorable subgroup is promoted to a primary result.

S2.2. *External background references*

Sobol and pycma CMA-ES retain their own search mechanisms and initialization. Their 1,296 runs and 302,400 paid evaluations passed separate journal audits. For descriptive common-scale comparison only, each external run's final error is divided by the matching internal common-initialization error, and medians are taken after this run-level normalization. The external optimizers never receive those internal labels. This supports budget-matched background comparison, not a same-history causal comparison or reproduction of published SAEA competitors.

Table S8. Descriptive background endpoint scores over all 24 functions. Lower is better; no additional test family is introduced.

D	Fixed .1	PEM-T	PEM-ST	Sobol	CMA-ES
10	-0.8823	-0.7156	-0.7465	-0.0805	-0.4275
20	-0.8761	-0.7763	-0.7877	-0.1066	-0.4799
40	-0.7934	-0.7295	-0.7041	-0.1029	-0.4440

GP-EI and author TuRBO use separate frozen protocols and strict pre-call budget guards. Their complete audited outcomes are now included in the main article. Author-code restart counterexamples are control-flow checks, not performance observations. Resource precheck instances are excluded from every benchmark table.

S3. Complete BBOB references and exclusive resource evidence

S3.1. *Full descriptive BBOB reference evidence*

Both frozen GP-EI and author TuRBO-1 batches complete 648 runs and 151,200 paid evaluations each, without reducing functions, repeats, GP training, or acquisition candidates. Together with the six internal and two basic references, all ten methods include 6,480 runs and 1,512,000 paid evaluations. External methods retain their own search and initialization. The matching internal initial-error denominator enters analysis only; normalization precedes within-instance medians. The 24 function scores are equally weighted. These descriptive comparisons add no significance family and do not reproduce the unavailable SAEA competitors.

GP-EI implements classical expected improvement (Jones, Schonlau, and Welch 1998) using the explicitly frozen kernel and search; it is not an exact reproduction of original EGO software. TuRBO-1 uses the author local trust-region implementation (Eriksson et al. 2019) with an external paid-call guard.

Table S9. Full ten-method function-mean log-relative endpoint scores. Lower is better.

Method	D=10	D=20	D=40
greedy	-0.9152	-0.9014	-0.8158
fixed01	-0.8823	-0.8761	-0.7934
fixed03	-0.7829	-0.771	-0.7097
pool_random	-0.1683	-0.2223	-0.2623
PEM_T	-0.7156	-0.7763	-0.7295
PEM_ST	-0.7465	-0.7877	-0.7041
sobol	-0.0805	-0.1066	-0.1029
cma_es	-0.4275	-0.4799	-0.444
gp_ei	-1.3292	-1.033	-0.9323
turbo_1	-1.102	-1.1219	-1.0389

The complete ten-method BBOB endpoint comparison is shown in Figure S5.

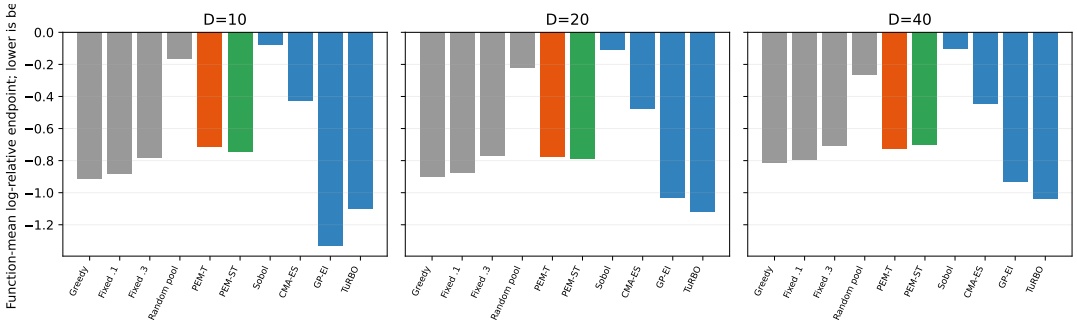


Figure S5. Complete BBOB background reference endpoints; no partial runs or resource-precheck instances.

Figure S5 retains the full ten-method ordering in each dimension. TuRBO-1 and GP-EI occupy different positions across dimensions, while Sobol remains close to zero and the internal methods form a tighter negative-error group. The panel-wise variation supports use of these methods as descriptive context; it does not justify a cross-protocol causal ranking because their initialization and search mechanisms differ.

S3.2. *Exclusive cost evidence*

The frozen resource design contains 54 fresh, serial process measurements on f1/instance4 across D=10/20/40 and three repeats of six methods. Their 12,600 paid calls are reported as cost observations, excluded from benchmark effect estimation. Actual process and background-CPU guards precede measurements. Runtime includes journaling and each method retains its own initialization. Greedy/fixed controls still perform temporal monitoring, so these are implemented-system costs, not stripped-down ideal algorithm costs.

Table S10. D=10; three-run medians; full min/max and per-process observations remain in the supplement CSV.

Method	Algorithm s	Nonobjective s	Peak GiB	Post-init prep s
PEM_ST	0.6547	0.6543	0.0945	0.0108
PEM_T	0.6075	0.6072	0.0947	0.01
fixed01	0.602	0.6016	0.0947	0.0099
gp_ei	27.6737	27.6724	0.2863	0.3456
greedy	0.5983	0.5979	0.0945	0.0098
turbo_1	7.4536	7.4522	0.3905	0.0923

Table S11. D=20; three-run medians; full min/max and per-process observations remain in the supplement CSV.

Method	Algorithm s	Nonobjective s	Peak GiB	Post-init prep s
PEM_ST	1.6909	1.6901	0.0946	0.012
PEM_T	1.5254	1.5246	0.0948	0.0108
fixed01	1.5252	1.5244	0.0948	0.0108
gp_ei	175.864	175.861	0.3059	1.0989
greedy	1.5246	1.5238	0.0945	0.0108
turbo_1	35.0159	35.013	0.5936	0.2183

Table S12. D=40; three-run medians; full min/max and per-process observations remain in the supplement CSV.

Method	Algorithm s	Nonobjective s	Peak GiB	Post-init prep s
PEM_ST	4.7496	4.7476	0.096	0.0157
PEM_T	4.3878	4.3856	0.0962	0.0145
fixed01	4.3787	4.3766	0.0965	0.0145
gp_ei	1308.81	1308.8	0.3073	4.0897
greedy	4.3615	4.3594	0.0958	0.0144
turbo_1	255.912	255.906	1.0791	0.7993

The measured-overhead and assumed-objective-cost scenarios are shown in Figure S6.

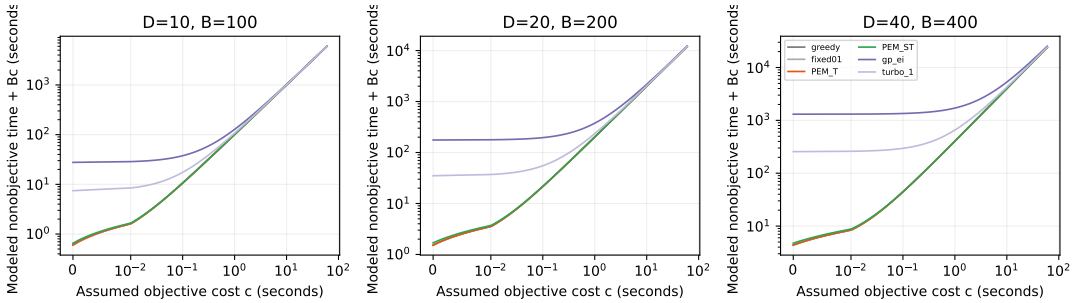


Figure S6. Hypothetical objective-cost scenarios from serial measurements; all methods use the same B in a dimension.

Figure S6 separates measured algorithm overhead from an assumed per-evaluation objective cost. The left side of each log-scale panel is dominated by method overhead, which is largest for GP-EI and TuRBO-1. As objective cost increases, all lines approach the common slope imposed by the identical evaluation budget. The intersections mark cost regimes in which overhead ceases to dominate; they are scenario calculations, not measured parallel speedups.

At an assumed objective cost c, total time is modeled as measured nonobjective time plus Bc. This scenario is not an observed application acceleration. Equal evaluation budgets have equal objective-cost slopes and cannot establish evaluation savings.

S4. LassoBench application transfer

S4.1. *LassoBench mechanism transfer*

LassoBench (Sehic et al. 2022) provides the application objectives. A separate protocol is frozen before formal seed42 labels. Three original tasks (D=60/100/180), eight methods, twenty optimization repeats, and 500 evaluations yield 480 runs and 240,000 full configuration evaluations (1,200,000 CV-fold solves). Internal initialization is 140/220/380 paid calls; external initialization remains native. The author five-fold full-fidelity objective, fixed data/splits, solver settings, bounds, and target definitions are unchanged.

Unknown optimum is never imputed. The main endpoint is original validation loss. After all 500 labels are saved, the first lowest-validation configuration receives exactly one author test call, adding 480 single fits separately. Synthetic test MSPE uses the author fixed oracle normalizer, while DNA uses absolute test MSE; these scales are not pooled. The synthetic constructor computes its fixed test normalizer, but no test-performance result enters any optimization feedback.

Table S13. synt_simple; twenty optimization-run dispersion and post-completion test medians.

Method	CV median	Q1	Q3	Test MSPE median	F-score median
PEM_ST	0.7896	0.7656	0.8528	1.1435	0.6667
PEM_T	0.7714	0.7557	0.8654	1.2013	0.6667
cma_es	0.758	0.7176	0.7826	1.072	0.5455
fixed01	0.7534	0.7092	0.816	1.1441	0.6667
gp_ei	1.0118	0.9392	1.0847	1.1167	0.6
greedy	0.7501	0.703	0.7755	1.1078	0.6
sobol	1.2057	1.0326	1.2544	1.1865	0.4808
turbo_1	0.6448	0.5888	0.7032	1.0643	0.6

Method	CV median	Q1	Q3	Test MSPE median	F-score median

Table S14. synt_medium; twenty optimization-run dispersion and post-completion test medians.

Method	CV median	Q1	Q3	Test MSPE median	F-score median
PEM_ST	1.146	1.1122	1.1731	0.7697	0.7692
PEM_T	1.1519	1.1327	1.1684	0.7562	0.8013
cma_es	1.1888	1.1345	1.237	0.7927	0.625
fixed01	1.1186	1.0751	1.1418	0.7486	0.7692
gp_ei	1.61	1.5226	1.7486	0.8469	0.6667
greedy	1.1103	1.069	1.1616	0.7703	0.7692
sobol	1.8953	1.7099	2.0302	0.8846	0.625
turbo_1	1.0277	0.9575	1.0632	0.835	0.625

Table S15. DNA; twenty optimization-run dispersion and post-completion test medians.

Method	CV median	Q1	Q3	Test MSPE median	F-score median
PEM_ST	0.3071	0.3056	0.3092	0.3304	NA
PEM_T	0.3073	0.3054	0.3093	0.3277	NA
cma_es	0.3014	0.2992	0.3043	0.3285	NA
fixed01	0.306	0.3025	0.3079	0.3307	NA
gp_ei	0.336	0.3328	0.3384	0.3495	NA
greedy	0.3057	0.3042	0.3067	0.327	NA
sobol	0.3348	0.332	0.3375	0.3439	NA
turbo_1	0.3028	0.3005	0.3052	0.3278	NA

Table S16. Prespecified application family, PEM-T minus fixed .1 raw final validation loss. Positive favors the fixed control. Intervals are unadjusted paired-run bootstrap.

Task	Median difference	CI low	CI high	Holm p	Loss	Win	Tie
synt_ simple	0.0503	0.0032	0.0725	0.0826	14	6	0
synt_ medium	0.0393	0.0221	0.0626	0.0192	16	4	0
DNA	0.0027	0.0005279	0.0044	0.0592	16	4	0

synt_simple: higher, Holm p=0.0826; synt_medium: higher, Holm p=0.0192; DNA: higher, Holm p=0.0592.

The three comparisons form an application-only Holm family, separate from the nine BBOB mechanism tests. Twenty repeats describe optimization randomness on one fixed dataset and split per task, not twenty independent datasets. The Wilcoxon and paired bootstrap assumptions and limits are retained regardless of significance or direction.

Original-scale validation endpoints for the three LassoBench tasks are shown in Figure S7.

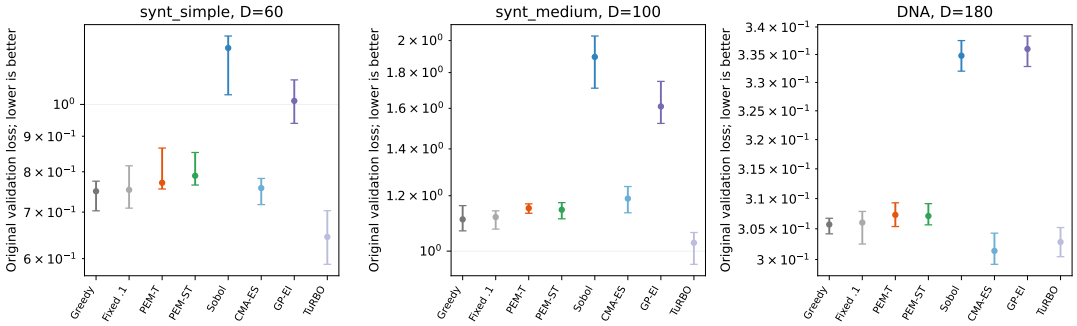


Figure S7. Application original validation endpoints with run-IQR bars; unknown-optimum error is never constructed.

Figure S7 reports the original validation loss on its native scale for each LassoBench task. The synt_simple and synt_medium panels separate the lower-loss internal and trust-region methods from Sobol and GP-EI, whereas the DNA panel compresses most methods into a narrow band. The run-interquartile bars reveal substantial dispersion for the weaker synthetic-task endpoints and discourage conclusions based only on median rank.

Their median and interquartile convergence histories are shown in Figure S8.

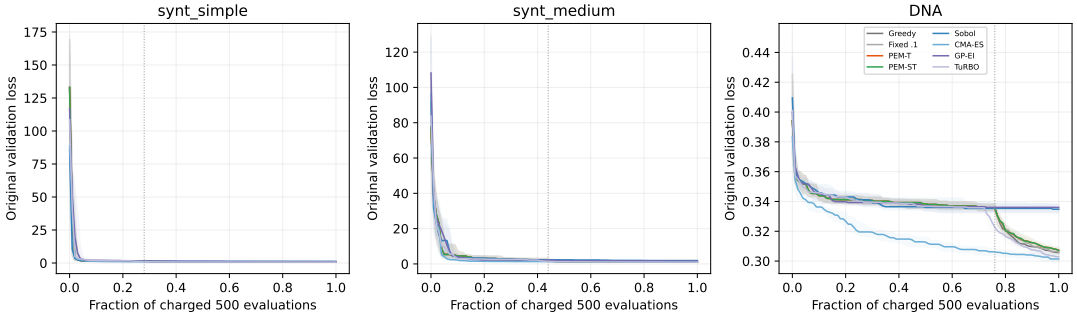


Figure S8. Application original-loss median and IQR curves. The dotted line marks internal initialization; external designs differ. Zero fraction displays the first paid label.

Figure S8 shows how the endpoint differences arise. Most improvement on the two synthetic tasks occurs early, followed by long plateaus, while DNA retains smaller and more gradual changes. The dotted initialization boundary applies to the internal configurations only; external curves begin from their native designs. The figure is therefore used to describe timing and stability, not to claim identical starting conditions.

Table S17. Descriptive post-initialization selected-point monitoring; no hidden pool objectives are acquired.

Task	Method	Failure	Q-loss	Claims	False	Pooled FDP	No-claim runs
DNA	PEM_ST	0.0704	0.0698	0	0	NA	20
DNA	PEM_T	0.0629	0.084	0	0	NA	20
DNA	fixed01	0.0346	0.0627	0	0	NA	20
DNA	greedy	0.0079	0.041	1	1	1.0000	19
synt_ medium	PEM_ST	0.1036	0.0778	53	46	0.8679	1
synt_ medium	PEM_T	0.0887	0.0904	7	6	0.8571	14

Task	Method	Failure	Q-loss	Claims	False	Pooled FDP	No-claim runs
synt_ medium	fixed01	0.0505	0.0452	27	23	0.8519	5
synt_ medium	greedy	0.0134	0.0147	60	54	0.9000	1
synt_ simple	PEM_ST	0.1103	0.0946	75	74	0.9867	1
synt_ simple	PEM_T	0.0931	0.1104	7	6	0.8571	15
synt_ simple	fixed01	0.0549	0.0476	76	74	0.9737	1
synt_ simple	greedy	0.0367	0.0137	220	218	0.9909	0

These closed-loop histories differ between methods and do not identify same-pool causal effects. Zero declarations give NA FDP, with counts disclosed. The 300-dimensional synt_ high extension therefore uses its separately frozen 1,000-evaluation protocol, reported in Section S5. Application findings cannot undo the failed original development gate.

S5. Predefined 300-dimensional extension

The predefined synt_high extension used 1,000 paid evaluations because the unchanged internal initialization requires 620, exceeding the 500-evaluation primary-application budget. Eight methods completed 20 optimization runs on one fixed seed42 data set and split (160 runs; 160,000 validation evaluations; 800,000 CV-fold solves), followed by exactly one held-out test call per completed run. The low-effective-dimensional synthetic construction and different budget preclude a general high-dimensional or cross-task superiority claim.

S5.1. *Endpoints and convergence*

Table S18. Endpoints and convergence.

Method	Validation median [Q1, Q3]	Test MSPE median [Q1, Q3]	Test F-score median [Q1, Q3]
greedy	1.38 [1.334, 1.435]	1.508 [1.466, 1.608]	0.6897 [0.6522, 0.7186]
fixed01	1.392 [1.345, 1.442]	1.615 [1.484, 1.677]	0.6977 [0.6596, 0.7317]
PEM_T	1.364 [1.338, 1.439]	1.58 [1.503, 1.653]	0.7143 [0.678, 0.7548]
PEM_ST	1.372 [1.312, 1.476]	1.614 [1.525, 1.653]	0.6977 [0.6818, 0.7317]
sobol	18.54 [16.12, 22.64]	13.45 [10.07, 19.15]	0.3314 [0.3055, 0.354]
cma_es	1.145 [1.089, 1.179]	1.696 [1.612, 1.909]	0.6522 [0.6, 0.6667]
gp_ei	17.04 [14.89, 21]	13.5 [9.747, 19.81]	0.359 [0.3123, 0.3709]
turbo_1	1.157 [1.128, 1.265]	1.654 [1.6, 1.798]	0.723 [0.6977, 0.75]

The 300-dimensional endpoint distribution is shown in Figure S9.

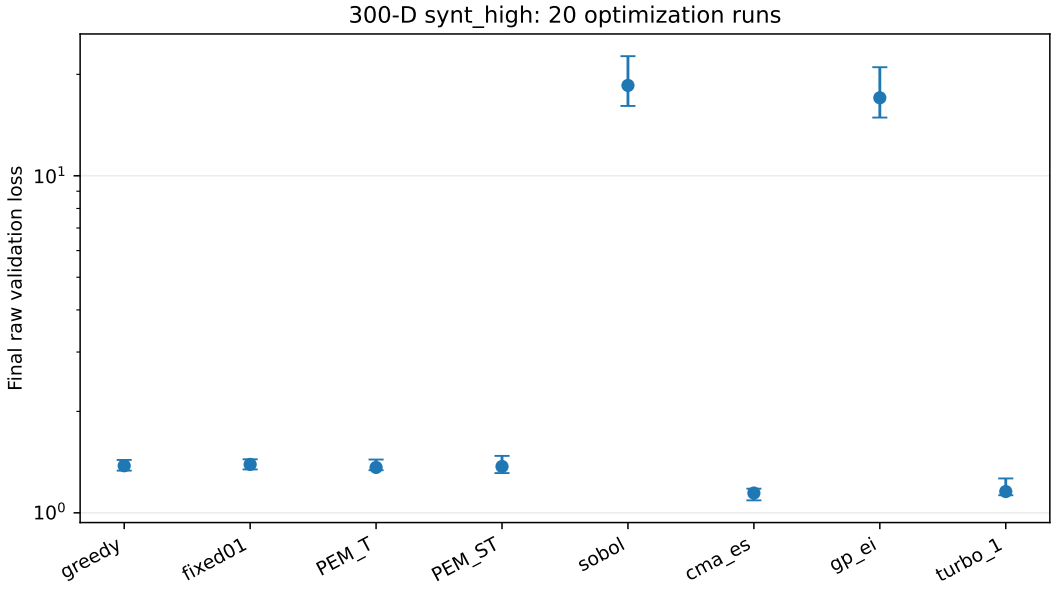


Figure S9. High-dimensional endpoints

Figure S9 separates the 300-dimensional methods into two endpoint groups. The four internal configurations, CMA-ES, and TuRBO-1 finish near the lower cluster, whereas Sobol and GP-EI remain more than an order of magnitude higher on the logarithmic axis. Interquartile bars are shown for all methods so that the apparent grouping can be assessed against between-run variability.

The associated paid-budget convergence histories are shown in Figure S10.

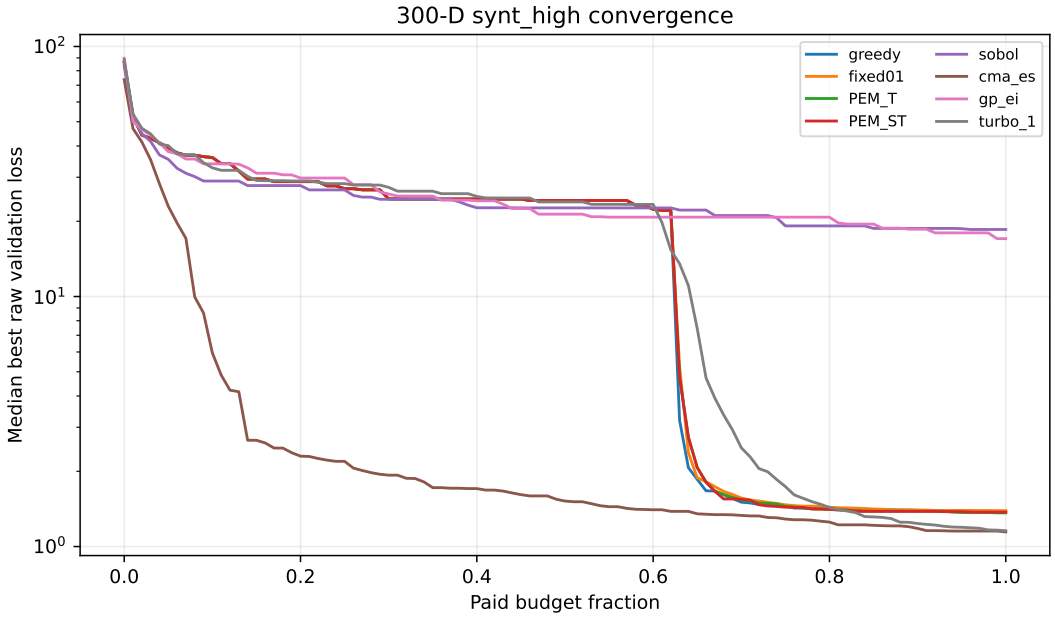


Figure S10. High-dimensional convergence

Figure S10 reveals distinct routes to the lower endpoint cluster. TuRBO-1 improves

early and continuously; the internal methods change sharply only after their long initialization consumes about 62% of the budget. CMA-ES improves later, while Sobol and GP-EI remain on the upper plateau. This timing difference is why the 1,000-evaluation protocol is reported separately from the primary 500-evaluation application study.

The prespecified descriptive PEM_T-minus-fixed 01 median raw-loss difference was -0.01788, with paired-run bootstrap interval [-0.06477, 0.05017] and 11/9/0 negative/positive/zero differences. No p-value was computed and this contrast was not added to the three-test primary application family. The interval conditions on one fixed data set and split.

S5.2. *Selected-point feedback and resource boundary*

Table S19. Selected-point feedback and resource boundary.

Method	Failure	QLoss	Explore	Fallback	Claims	FDP
greedy	0.07447	0.01695	0	0	257	1
fixed01	0.07895	0.03878	0.09618	0	127	0.9921
PEM_T	0.08921	0.07535	0.3405	0	5	1
PEM_ST	0.1266	0.06124	0.3854	0.01184	119	0.9916

Feedback summaries cover the 380 post-initialization selections per internal run. They describe each closed-loop history and do not establish same-pool causal effects or conformal error control. Before formal labels, eight separate full-budget resource runs completed 8,000 validation evaluations and zero test calls. Their mixed-load timings supported feasibility only; formal optimization times are reported as observed process costs rather than exclusive speedups.

S6. Prespecified model and training-cap sensitivity

Five preselected BBOB structure representatives (f1, f6, f10, f15, f20), dimensions 20/40, instances 14–16, and three optimization repeats evaluated PEM_T under the base setting and four one-factor perturbations. All 450 runs and 135,000 paid evaluations completed; no hidden diagnostic labels or inferential tests were added. This local sensitivity study does not select a replacement default.

Table S20. Prespecified model and training-cap sensitivity.

D	Variant	Endpoint median [Q1, Q3]	AUC median [Q1, Q3]
20	base	-1.536 [-2.168, -0.8568]	-0.962 [-1.292, -0.5037]
20	cap128	-1.439 [-1.903, -0.7494]	-0.9267 [-1.28, -0.5181]
20	cap512	-1.536 [-2.168, -0.8568]	-0.962 [-1.292, -0.5037]
20	scale125	-1.085 [-1.722, -0.6441]	-0.6593 [-1.087, -0.3838]
20	scale500	-1.473 [-2.095, -0.6943]	-0.776 [-1.079, -0.4353]
40	base	-1.275 [-1.676, -0.8544]	-0.8335 [-1.096, -0.5196]
40	cap128	-0.9176 [-1.179, -0.5716]	-0.6855 [-0.8808, -0.4435]
40	cap512	-1.235 [-1.69, -0.8008]	-0.8185 [-1.123, -0.5374]
40	scale125	-0.8456 [-1.18, -0.5364]	-0.5406 [-0.7806, -0.3901]

D	Variant	Endpoint median [Q1, Q3]	AUC median [Q1, Q3]
40	scale500	-1.151 [-1.311, -0.7564]	-0.7301 [-0.9111, -0.4361]

Paired model and training-cap endpoint distributions are shown in Figure S11.

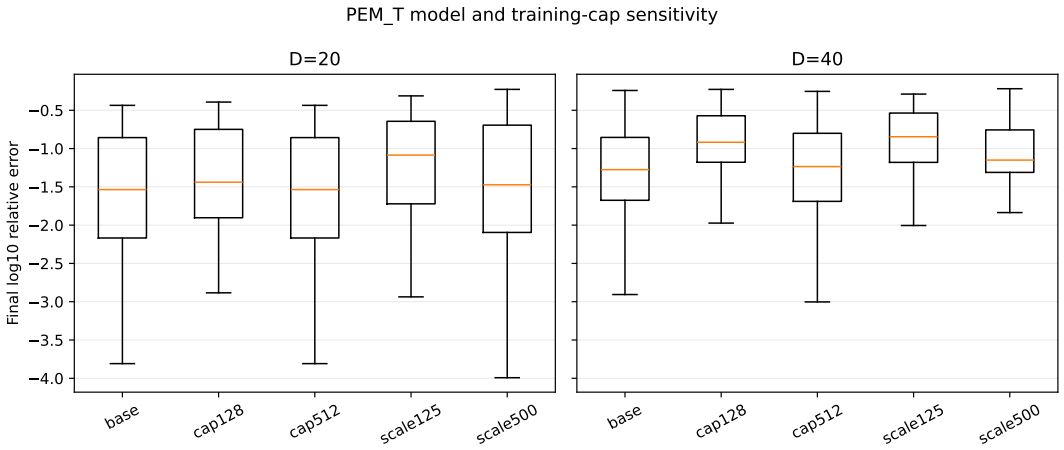


Figure S11. Model and training-cap sensitivity

Figure S11 displays paired endpoint distributions for the base model, two training caps, and two length-scale variants. Cap512 coincides with the base configuration in dimension 20 because the cap is never reached. Other variants shift the median and spread, especially in dimension 40, but none converts the sensitivity analysis into an unqualified favorable result for the feedback mechanism.

Paired differences below are variant minus base in final log10 relative error. Negative values favor the variant descriptively.

Table S21. Prespecified model and training-cap sensitivity.

D	Variant	Difference median [Q1, Q3]	Negative	Positive	Zero
20	cap128	0.06854 [0, 0.1836]	11	30	4
20	cap512	0 [0, 0]	0	0	45
20	scale125	0.2805 [0.08773, 0.5608]	8	37	0
20	scale500	0.09344 [-0.04563, 0.2851]	14	31	0
40	cap128	0.3284 [0.0957, 0.4718]	4	41	0
40	cap512	0.005203 [-0.02706, 0.08161]	18	27	0
40	scale125	0.3253 [0.03915, 0.656]	8	37	0
40	scale500	0.09029 [-0.002699, 0.2816]	12	33	0

The 45 D=20 base/cap512 pairs had identical PRE/LABEL trajectories and candidate arrays, as neither cap was reached. Thus that equality is an implementation invariant rather than evidence that caps never matter. At D=40, cap512 is untruncated only within this 400-evaluation scope.

Table S22. Prespecified model and training-cap sensitivity.

D	Variant	Capped	Failure	Explore	Fallback
20	base	0	0.08746	0.2789	0
20	cap128	3195	0.1332	0.3444	0
20	cap512	0	0.08746	0.2789	0
20	scale125	0	0.1002	0.3173	0
20	scale500	0	0.07175	0.283	0
40	base	6435	0.08889	0.2541	0
40	cap128	12195	0.1893	0.5487	0
40	cap512	0	0.08993	0.2591	0
40	scale125	6435	0.09733	0.2797	0
40	scale500	6435	0.08674	0.2902	0

Feedback metrics follow each variant's distinct closed-loop history. The limited function/instance set, shared initialization, and absence of multiplicity-adjusted inference restrict these results to robustness description.

S7. Prespecified budget-performance sensitivity and recovery accounting

S7.1. *Design and aggregation*

Five prespecified BBOB structure representatives (f1, f6, f10, f15, and f20), dimensions 20/40, instances 17–19, and three repeats were run independently at total budgets 5D, 10D, and 15D. Six internal configurations and four external references completed all 2,700 runs and 810,000 paid evaluations, with no diagnostic labels. Each endpoint uses the matching fixed 01 common-initialization error as a post-run denominator. The repeat median is first taken within each function-instance cell; the three instances within each function and the five function scores are then averaged with equal weight. No hypothesis test, confidence interval, or parameter selection was added.

S7.2. *Endpoints and convergence*

Lower values are better. Each table entry is the prespecified hierarchical mean final log10 relative error.

Table S23. Endpoints and convergence.

Method	D20-5D	D20-10D	D20-15D	D40-5D	D40-10D	D40-15D
greedy	-0.961	-1.768	-2.097	-0.9999	-1.544	-1.643
fixed01	-0.9322	-1.749	-2.035	-0.9727	-1.53	-1.664
fixed03	-0.6895	-1.616	-2.041	-0.7998	-1.491	-1.634
pool_random	-0.08979	-0.3369	-0.433	-0.187	-0.3509	-0.4963
PEM_T	-0.7538	-1.689	-1.987	-0.8891	-1.477	-1.598
PEM_ST	-0.7407	-1.676	-2.039	-0.8906	-1.456	-1.57
sobol	-0.1024	-0.1513	-0.209	-0.0429	-0.09157	-0.09792

Method	D20-5D	D20-10D	D20-15D	D40-5D	D40-10D	D40-15D
cma_es	-0.4835	-0.7414	-1.068	-0.4553	-0.6883	-0.949
gp_ei	-1.642	-2.388	-2.574	-1.651	-1.993	-2.174
turbo_1	-1.08	-2.216	-2.502	-0.9778	-1.943	-2.516

Endpoint changes across the three prespecified total budgets are shown in Figure S12.

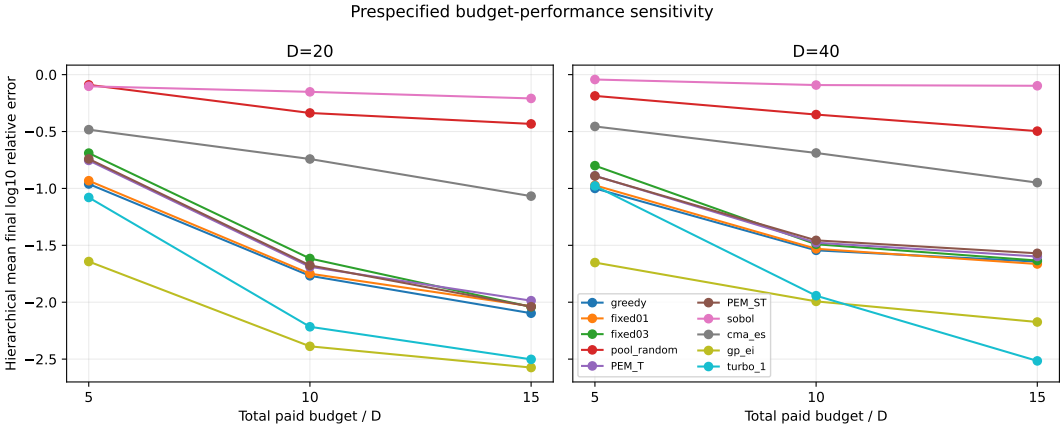


Figure S12. Budget-sensitivity endpoints

Figure S12 connects endpoint quality to total budget. Most methods improve as the allowance rises from 5D to 15D, but the monitored internal variants remain above the strongest external curves. The slopes differ by dimension, which indicates that additional budget changes the magnitude of the gap without reversing the principal mechanism conclusion.

The corresponding independent-run convergence histories are shown in Figure S13.

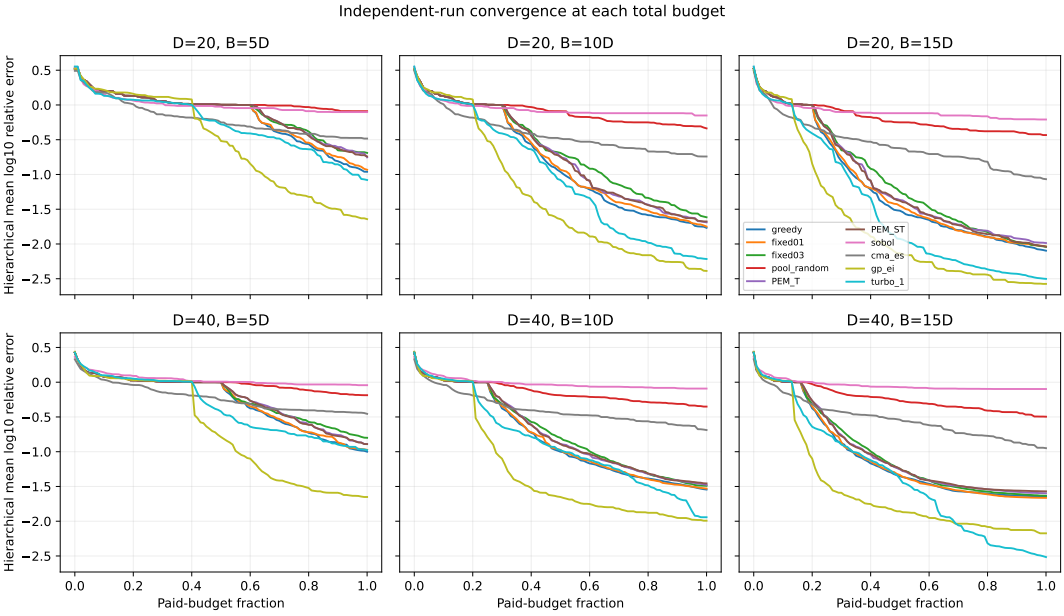


Figure S13. Budget-sensitivity convergence

Figure S13 plots each independent run against its own paid-budget fraction. The 5D, 10D, and 15D panels share early behavior where the information prefixes are equal, then separate as longer runs acquire additional labels. External methods do not share those internal prefixes. The six panels therefore expose both convergence shape and the information-order boundary used in the recovery audit.

Each budget is a separate optimization run. A longer-budget endpoint therefore supplies robustness evidence under a larger total allowance; it is not a continuation selected after observing the shorter run. The restricted five-function, two-dimension design and descriptive aggregation do not establish a general ranking.

S7.3. *Prefix and information-order audit*

Table S24. Prefix and information-order audit.

Method	Equal 5D/10D/15D prefixes	Required invariant
greedy	90/90	yes
fixed01	90/90	yes
fixed03	90/90	yes
pool_random	90/90	yes
PEM_T	90/90	yes
PEM_ST	90/90	yes
sobol	90/90	no
cma_es	90/90	no
gp_ei	0/90	no
turbo_1	0/90	no

The budget-blind internal core was required to reproduce identical 5D and 10D prefixes inside the corresponding longer run; all 540 required comparisons passed. External prefix equality was recorded rather than required: 180/360 comparisons were equal. That count is an implementation observation, not an assumption used to validate external methods. Full audits replayed all 810,000 PRE/LABEL pairs, internal bounds and management states, and 7,020 raw-file hashes. No total-budget value entered the internal proposal rule.

S7.4. *Recovery accounting and evidence boundary*

The budget-sensitivity interruption left 8 incomplete units with 1,791 physically acquired labels. Their complete directories were archived before resumption, and each affected unit was rerun from its frozen seed. These labels never entered the 2,700 canonical trajectories. The budget-sensitivity physical ledger is therefore 811,791 calls: 810,000 canonical evaluations plus 1,791 disclosed interrupted-run calls. The additional calls are excluded from endpoints, convergence curves, and algorithm budgets.

The completed GP-EI performance batch contains 151,200 canonical algorithm evaluations. During interruption recovery, one GP-EI point (evaluation 349 of gp_ei_f02_d40_i13_r2_b400) was physically queried a second time because the newer partial journal contained 348 labels while an earlier archive contained 349. The point and returned label matched exactly; the label entered the optimizer history once, so the canonical 400-step trajectory and endpoint are unchanged. The physical GP-EI ledger is therefore 151,201 calls: 151,200 canonical evaluations plus one disclosed recovery-validation call. The extra call is excluded from algorithm budgets, performance curves, and exclusive-cost analysis.

The read-only recovery audit made zero objective calls and verified eight recovered trajectories (2,800 canonical evaluations), 2,014 saved second-generation labels, and 786 second-generation physical calls, one of which overlaps the earlier archive.

This section does not reverse the failed development benefit gate. The paper is positioned as a decision-consequence auditing framework and makes no general optimizer-superiority claim. Full CEC confirmation is not required for that framework scope. A CEC batch started on 24 September 2026 was stopped after the scope decision; its 30 complete and 8 partial trajectories are retained as exploratory records and excluded from manuscript statistics. A verifiable author-code DRSO comparison is provided in S8.2; the three originally named but unavailable implementations continue to delimit method-specific claims.

S8. Reproducibility inventory

The archived evidence contains frozen protocols, source hashes, dependency locks, deterministic seed maps, prediction-before-label journals, candidate arrays, separate diagnostic labels, endpoint re-evaluations, raw-file hashes, analysis tables, and figure sources. Diagnostic labels never enter optimizer histories. Interrupted or repeated physical calls are disclosed separately from canonical algorithm budgets. The repository contains sources and curated revision evidence. The byte-preserving GP source bundle, frozen intervention records and lossless compact transfer release support the added results. Larger historical archives remain available from the corresponding author upon reasonable request. Frozen source revisions, data-release commits and file checksums identify the evidence.

S8.1. *Finite-element audit*

The EngiBench beam batch contains 50 complete runs, 10,000 canonical finite-element evaluations, 50 independent endpoint re-evaluations, zero failures, and maximum volume-fraction error 1.665e-16. The audit verifies the complete design, source hashes, 100 raw-file hashes, prediction-before-label ordering, exact endpoint replay, and separation of the earlier smoke run from method selection.

S8.2. *Matched-initialization external-reference, acquisition, and SAEA sensitivity*

The supplemental beam experiment gives GP-EI and TuRBO-1 exactly the same 58-point initial finite-element archive used by Greedy, fixed 01, and PEM-T in each repeat. It contains 20 complete trajectories and 4,000 algorithm-budget observations: 1,160 previously paid deterministic initial labels and 2,840 additional finite-element calls. The event-level audit finds zero failures, zero exact duplicate points, complete prediction-before-label order, and exact agreement with all ten source-archive hashes. Forty additional physical calls belong only to two independent endpoint-diagnostic rounds and never enter an optimizer history. The accepted diagnostic table has maximum absolute difference 1.50e-8 and maximum relative difference 6.23e-12.

Table S25. Matched-initialization external-reference, acquisition, and SAEA sensitivity.

Method	Shared-init median	Native-init median	Median shared-minus-native	Holm p	Median shared-minus-PEM-T	Wins/ties/ losses vs PEM-T	Holm p
GP-EI	2308.5	2117.5	+399.8	0.320	+242.8	1/3/6	0.0469
TuRBO-1	574.0	578.2	-44.7	0.695	-1372.3	10/0/0	0.00391

The initial launch stopped after a missing fixed-seed argument prevented two GP-EI workers from entering GP search; those logs contain only reused initial labels and zero subsequent objective calls. Six already-running TuRBO-1 units completed through an unaffected branch. The complete launch state was archived, the one-argument wiring error was corrected, and the remaining units resumed. The terminal audit accepts six original and fourteen corrected code hashes after verifying that the change lies outside the completed TuRBO branch. No failed or interrupted GP label enters the 20 canonical trajectories.

Two further Gaussian-process acquisition controls use the same ten 58-point archives, 200-evaluation budget, Matérn-5/2 model, 4,096-point Sobol candidate pool, and eight local optimization starts. GP-PI uses an improvement margin equal to 0.01 times the current response standard deviation. GP-UCB is implemented for minimization by selecting the lowest value of mean minus 2.0 standard deviations. These 20 trajectories contain 4,000 algorithm-budget observations, including 1,160 reused labels and 2,840 new finite-element calls. The audit verifies sequential prediction-before-label records, bounds, finite outputs, exact source-archive hashes, complete budgets, and zero exact duplicate points.

Table S25a. Additional shared-initialization acquisition and relation-model controls.

Method	Shared-init median	Q1	Q3	Median improvement after initialization	Median difference vs PEM-T	Wins/ties/ losses vs PEM-T	Holm p
GP-PI	1233.1	921.2	1399.7	0.517	-876.1	9/0/1	0.0117
GP-UCB	2308.7	2223.8	2827.3	0.000	+574.6	2/2/6	0.0391
DRSO	647.4	580.4	780.0	0.747	-1334.6	9/0/1	0.0117

The different outcomes under a common surrogate and common starting archive isolate acquisition behavior. Candidate-dependent uncertainty is structurally capable of changing decisions, but its practical value depends on how uncertainty and predicted objective value are combined. GP-PI supplies a successful candidate-dependent control; the fixed-multiplier GP-UCB control does not improve the median initial incumbent under this budget. The three rows form one Holm family.

DRSO provides the requested competitive surrogate-assisted evolutionary comparison. The public author repository is frozen at commit `03c19d4280b38230e0852054922072f5aa6dcdf5` under LGPL-3.0. Its relation models, unevaluated-population mechanism, reproduction, training cap of 100, and other algorithm settings remain unchanged. Population size is set from the author default of 50 to 58 only because the common archive contains 58 points. A Python 3.12 compatibility namespace supplies the removed `imp` name required by pyDOE2 and an unused optional callback import; neither compatibility file alters the DRSO search implementation.

The DRSO batch contains ten complete trajectories and 2,000 algorithm-budget observations: 580 reused initial labels and 1,420 new finite-element calls. All 4,010 event records have sequential PRE/LABEL order, all ten source-archive and author-tree hashes match, and no trajectory contains an out-of-bounds, nonfinite, or exact duplicate point. The author-code result object, external paid-call journal, and endpoint minimum agree in every repeat. DRSO improves median compliance by 74.7% after initialization and reaches 647.4, compared with 1972.8 for PEM-T.

S8.3. *Evidence boundary*

The empirical residual monitor has no claimed pointwise coverage theorem, false-discovery-rate guarantee, regret bound, or global convergence theorem. Broad BBOB comparisons with external methods retain native initialization and remain descriptive; the beam sensitivity alone supplies a matched initial archive. GP-PI and GP-UCB are transparent acquisition controls, while DRSO is an author-code published SAEA. The 19-dimensional beam family does not establish performance for unrestricted 5,000-variable topology optimization. Full CEC exploration was stopped after the decision-consequence auditing scope was selected; its exploratory trajectories are excluded from all article statistics. Temporal CP-SAEA, SADE-SS, and BGP-SAEA remain excluded because matching executable author implementations with frozen versions, parameters, and licenses were unavailable; similarly named public implementations are not substituted for them. DRSO closes the requested competitive-SAEA evidence category without claiming to reproduce those unavailable methods.

S9. Frozen GP intervention and immediate diagnostic accounting

This additional study is separately frozen, descriptive, and not part of the original Holm families. Input hashes and 120 decisions were committed at f1c9bbdebc35071bd935d0b11c92857f5e07201e before new intervention labels. The initially transmitted binary ZIP was incomplete, despite an intact local author-supplied archive. Local scientific input hashes and frozen choices did not change. The public input release now uses a byte-preserving text bundle at commit 71e074945262e1dbfc223726c7216e5415385ddc, replacing the incomplete repository binary. Its restoration checks every original file hash, all segment hashes and the decoded payload hash before writing any file. It preserves all 60 original file bytes, including line endings; it does not claim to reconstruct the original ZIP container bytes. The original supplied archive remains identified by SHA256 c59079aaf34f560ee2f3e93a55bd60e13b2fa4c53097285842580df4d3b8098b. Public distribution integrity and scientific execution integrity are separate checks.

Thirty source histories contain 6,000 budgeted algorithm observations and 580 distinct common initial labels across ten repeats. Frozen states stop at PRE(t), exclude LABEL(t), and contain only observations before 59,101,151 or 200. The same recorded ARD Matérn-5/2 posterior scores mean, EI, PI and LCB on 4,096 scrambled Sobol points plus the original action. EI uses zero margin, PI uses 0.01 times the training standard deviation, and LCB uses kappa 2. Logged hyperparameters are not refitted. All120 source-action mean/standard-deviation comparisons satisfy rtol=atol=1e-8. Decisions and candidate records are hashed before source action labels are looked up. No new diagnostic label enters a source optimizer history.

Table S26. All controls and source-history groups. Four checkpoint estimates are averaged within each repeat before averaging ten repeats. U and DeltaU are in units of 1e-6.

Source history	Control	Changed (%)	Mean U (1e-6)	DeltaU vs mean (1e-6)
EI history	MEAN	0.0	0.899669	+0.000000
EI history	EI	30.0	0.899669	+0.000000
EI history	PI	25.0	0.899669	+0.000000
EI history	LCB	52.5	0.000000	-0.899669
PI history	MEAN	0.0	0.347088	+0.000000
PI history	EI	95.0	0.000000	-0.347088
PI history	PI	7.5	0.347088	+0.000000
PI history	LCB	97.5	0.000000	-0.347088
UCB history	MEAN	0.0	0.877758	+0.000000
UCB history	EI	30.0	0.887232	+0.009475
UCB history	PI	35.0	1.178882	+0.301124
UCB history	LCB	75.0	0.511893	-0.365864

The training scale ranges from 733152 to 9.12154e+08. This can make normalized gains small even when raw compliance changes. The metric is retained as frozen; no post hoc replacement or new p-value is used. These comparisons neither test delayed exploration nor authorize promotion.

The 480 control-state choices reduce to 261 distinct within-state actions. Source-action reuse accounts for 120 responses; exact diagnostic reuse across states accounts for 3;138 are new FE calls. New diagnostic FE time is 4.437432s and GP state preparation time13.435883s. The initial solver compatibility check made42 calls before detecting a 4.929e-8 relative discrepancy at a large-compliance point. That discrepancy was numerical, not a failed FE response. The complete retained60-call check uses the disclosed1e-6 relative tolerance and passes every comparison. The original prediction replay tolerance remains1e-8. The first42 outputs and timings were not retained but their attempts are included. The retained solver check takes 2.327726s. Total additional attempts for this GP revision are 240. These observed times do not establish exclusive wall-time savings.

S10. Independent load-position transfer, horizon effects and paid workflows

The protocol and executable implementation were committed at 7555d58b145a2c44b73bab5ec47d0f643a57dde1 before acquiring new transfer label-s. Conditions forcedist0.25 and 0.75 each have ten independently seeded58-point Sobol initializations. Fixed seed namespaces distinguish initialization, pilots and validation. The same case/step candidate namespace is shared across policies inside a phase. All policies use one shared specification; tie ordering, fit limits and budgets were not tuned using outcomes. This is within-family physical-condition transfer, not cross-domain validation.

Table S27. Differences between the two additional GP experiments. The original beam wrapper, material, supports, volume 0.35 and dimension 19 are retained.

Setting	Frozen intervention (S9)	Independent transfer (S10)
Source state	Archived EI/PI/UCB histories	Fresh initialization in each force condition
Kernel	Logged ARD Matérn-5/2	Isotropic Matérn-5/2

Setting	Frozen intervention (S9)	Independent transfer (S10)
Hyperparameters	Frozen, optimizer disabled	L-BFGS-B, max100 iterations, no restarts
Candidate pool	4096 Sobol + original action	1024 Sobol; observed points excluded
EI margin	0	0.01 times training standard deviation
PI margin / LCB kappa	0.01 scaled / 2	0.01 scaled / 2
GP alpha / normalization	1e-6 / normalized targets	1e-6 / normalized targets
Physical force position	0.0	0.25 and 0.75

The transfer constant bounds are[0.05,20] and length-scale bounds[0.005,2], with initial length scale 0.2. Fit convergence warnings are recorded and do not trigger restarts. Beams2D caches force-dependent matrices; each case creates a fresh problem object configured once with its actual force position. This prevents cross-condition state reuse. The environment pins NumPy 2.2.6, SciPy 1.15.3, scikit-learn1.7.2, EngiBench 0.2.0 and cvxopt1.3.2. These settings define a separate controlled study and do not rerun the historical ARD/continuous-refinement GP optimizer.

S10.1. *Paid selection and independent validation*

Each audited workflow pays 58 initial labels,4x20 pilot labels and 142 selected-policy validation labels:280. Each complete portfolio independently pays 58 initial labels and 4x142 continuation labels:626. A fixed single policy needs200. The pilot winner is the lowest final pilot compliance, with exact ties broken in PI, mean, EI, LCB order. Selection is persisted after 138 calls and before any canonical validation label. Canonical histories start again from the initial58 labels and never include pilot responses. Validation candidates use a different namespace; no warm-starting selector is claimed. The selected canonical trajectory and its separately paid portfolio counterpart agree exactly in every repeat. All duplicate designs across workflows are physically evaluated and counted, rather than credited as free labels.

Table S28. All twenty paired endpoints. The selected-policy column is the prespecified independent canonical endpoint. All-paid best is a secondary descriptive endpoint including every pilot design; it does not replace the validation criterion. Gap is 100x(selected/portfolio-best-1).

Force	Rep	Selected	Initial	Selected-policy	Portfolio-best	Gap (%)	All-paid best
0.25	0	PI	1559.326	1350.207	1350.207	0.000	1350.207
0.25	1	EI	1657.740	1192.164	805.832	47.942	1192.164
0.25	2	PI	1817.963	929.539	929.539	0.000	929.539
0.25	3	EI	3373.312	1048.640	923.095	13.600	1048.640
0.25	4	PI	1392.373	1152.823	1015.625	13.509	1152.823
0.25	5	PI	1426.446	1426.446	1195.130	19.355	1426.446
0.25	6	PI	2081.842	956.490	915.138	4.519	956.490
0.25	7	MEAN	1458.044	1204.446	1204.446	0.000	1065.517
0.25	8	PI	1189.521	956.887	956.887	0.000	956.887
0.25	9	MEAN	1774.608	1535.851	838.957	83.067	1392.803
0.75	0	PI	1287.356	430.374	430.374	0.000	430.374
0.75	1	PI	779.191	303.745	303.745	0.000	303.745
0.75	2	LCB	653.215	454.896	315.137	44.349	375.564
0.75	3	PI	703.816	326.950	326.950	0.000	326.950
0.75	4	MEAN	607.672	481.780	351.951	36.889	327.563

Force	Rep	Selected	Initial	Selected-policy	Portfolio-best	Gap (%)	All-paid best
0.75	5	PI	644.397	335.272	332.477	0.841	312.168
0.75	6	PI	697.634	383.191	325.992	17.546	383.191
0.75	7	PI	997.734	677.303	174.014	289.223	597.821
0.75	8	PI	583.133	429.673	274.350	56.615	285.959
0.75	9	PI	720.317	303.082	211.908	43.025	253.360

The primary median paired gaps are 9.013698% and 27.217435%. Exact portfolio-best matches are 4/10 and 3/10. The selected counts are PI6, EI2, mean2 at force 0.25, and PI8, LCB1, mean1 at force 0.75. Exact pilot-end ties occur in 7 and 4 repeats; counts should not be interpreted independently of the tie rule. The secondary all-paid gaps have medians9.013698% and 2.115716%. The discrepancy between primary and secondary endpoints is disclosed because the practical availability of a pilot design differs from evidence that the chosen policy transfers well.

Table S29. Each200-call fixed policy compared with the280-call selected workflow. Median effects are paired100x(selected/fixed-1); negative favors selection. Wins/ties/losses are descriptive, not tests of superiority or equivalence.

Force	Fixed policy	Median fixed compliance	Paired effect (%)	Selection wins/ties/losses
0.25	MEAN	1102.310	+0.000	4/2/4
0.25	EI	1233.521	-1.813	5/3/2
0.25	PI	956.689	+0.000	1/6/3
0.25	LCB	1360.425	-8.978	7/1/2
0.75	MEAN	326.834	+0.420	3/2/5
0.75	EI	379.451	-5.682	5/1/4
0.75	PI	359.232	+0.000	1/8/1
0.75	LCB	475.896	-6.764	5/3/2

Selecting the best fixed policy retrospectively from these validation outcomes would be a separate selection procedure. No such200-call oracle advantage is attributed to the audit or its comparator.

S10.2. *Horizon separation*

Table S30. Pilot improvement from the common initial incumbent. First and 20-step columns are medians of per-repeat percentages. The final count concerns the same pilot trajectory.

Force	Policy	First step (%)	20 steps (%)	Zero first, positive20 /10
0.25	MEAN	0.000	13.006	6
0.25	EI	0.000	9.706	4
0.25	PI	0.000	8.592	5
0.25	LCB	0.000	2.514	4
0.75	MEAN	0.000	42.350	8
0.75	EI	0.000	13.491	5
0.75	PI	0.000	46.765	7
0.75	LCB	0.000	3.649	4

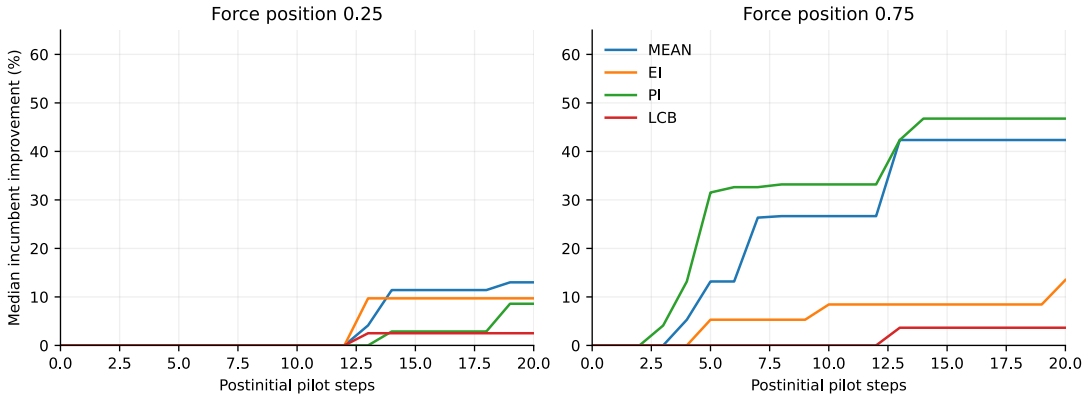


Figure S14. Pilot improvement across twenty steps under the two force positions.

Figure S14 plots condition-specific median incumbent improvement over the pilot horizon. It illustrates why a first action without gain is not a sufficient rejection rule. These are whole-policy trajectories with diverging histories; the figure does not estimate a first-action treatment effect with an identical continuation. Pilot-to-canonical selection transfer is a further question because the canonical candidate stream is independent.

S10.3. *Costs, recovery and evidence integrity*

Table S31. Additional FE-attempt boundary of the earlier GP/transfer stages. Later released checks are accounted for separately in Table S40; historical experiments before this revision are not included.

Component	Physical attempts	Use in reported endpoints
New frozen GP intervention	138	Separate diagnostics only
Solver compatibility verification	102	Never optimizer training
Earlier complete transfer execution	18120	Prior checked endpoints; raw journals unavailable in current workspace
Retained unchanged-protocol transfer recovery	18120	Retained canonical and pilot records, labels used once
Total including recovery	36480	No repeated attempt is hidden in a budget200 trajectory

The prior transfer journals became unavailable in the working environment before their raw release. An explicit recovery record at 62560a245a8b804dcff3ce00aa45aae0af9f3884 preserves the unchanged scientific configuration and caps additional recovery calls at 18,120. All twenty previously recorded selected-policy and portfolio-best endpoints are reproduced. Recovery contributes additional physical cost, not additional independent repeats or a changed inferential family. The initial zero-call bootstrap failure from the earlier launch adds no FE cost. The retained20 cases comprise2,320 physical initialization calls,1,600 pilot calls and 14,200 canonical postinitial calls. Their 100 full-budget trajectories contain 20,000 algorithm observations because initialization is shared among forks inside each workflow.

Table S32. Retained observed component times summed across ten repeats per condition, in seconds. Four workers run concurrently; sums are not exclusive elapsed-time benchmarks.

Force	Audited FE	Portfolio FE	Audited GP/selection	Portfolio GP/selection
0.25	90.395	194.651	57.242	175.964
0.75	87.050	190.999	53.674	171.385

The analyzer validates complete PRE/LABEL pairing,906 physical calls per case,280/626 workflow counts,20 pilot steps and 142 canonical steps per policy, selection before validation, correct seed namespaces, identical initial history hashes, exclusion of pilot labels from every canonical history, exact selected/portfolio trajectory replay, and every endpoint and component-time sum. It makes zero new FE calls. The compact release regenerates deterministic Sobol points and reconstructs each retained source JSONL byte for byte, checked against its SHA256; no objective call is needed to restore or analyze the records. Original full journals remain the source of timing and information-boundary checks.

There are no added p-values, confidence guarantees, retrospective deployment tolerances, claims of cross-domain transfer, or demonstrated production wall-time savings. The two loads strengthen within-family applicability. They do not establish optimizer superiority, a competitive selection advance over warm-starting, or validation of unrestricted5000-variable topology optimization.

Table S33. Settings of separate evidence releases. These implementations differ and are not pooled as exact replications. All new GP records use alpha 1e-6 and normalized targets.

Record	GP fitting	Candidate pool	Target
Original GP audit	Logged ARD Matérn-5/2; snapshot does not reoptimize	4096 plus source action	120 states; immediate alternatives
Common continuation	Same source ARD theta; fixed during branch updates	First: original pool; followup 1024	30 archived states; h0/1/3/5; mean continuation
Beam transfer	Isotropic Matérn-5/2; bounded L-BFGS-B, max 100 iterations per fit	1024	20 fresh cases; full-policy pilots/continuations
Independent budget	ARD theta fitted once at shared 50-label warm state, max 30 iterations; then fixed	1024; 8 calibration probes	10 new seeds; 100-label states; mean/EI followup 5
Thermal verification	Isotropic Matérn-5/2; fixed length 0.4; no hyperparameter fitting	256	12 variables; 24 initial labels; 96 calls/workflow

Table S34. Archived common-continuation denominators. Original 90-contrast reporting is retained; action equivalence supplies a supplementary denominator, not extra independent samples.

Reporting object	Count
Original named contrasts	90
Baseline-action aliases	34
Different-action named contrasts	56
Distinct nonbaseline-action contrasts	46
Named h0/h5 changes	8
Distinct-action changes	7
Distinct delayed non-ties	6
States containing changes	4

Table S35. Archived first-action contrasts with common posterior-mean continuation. Better/tie/worse counts use the original named-control denominator; each row has 10 dependent-source states and margin 1e-6. Positive advantage favours the named control.

Source history	Control	h0 B/T/W	h5 B/T/W	Median h5 (%)
EI	EI	0/10/0	1/9/0	0.000
EI	PI	0/10/0	1/8/1	0.000
EI	LCB	0/10/0	1/9/0	0.000
PI	EI	0/10/0	0/10/0	0.000
PI	PI	0/10/0	0/10/0	0.000
PI	LCB	0/10/0	0/10/0	0.000
UCB	EI	1/7/2	0/7/3	0.000
UCB	PI	0/9/1	0/8/2	0.000
UCB	LCB	0/8/2	1/6/3	0.000

Table S36. Independent budget replay: horizon 5, margin 0.001,25 distinct nonbaseline-action contrasts per continuation. Requests are conservatively charged response lookups across 30 states. Calibration is a constructed root-EI proxy gate, not a published diagnostic algorithm. Completed-horizon zero errors come from measuring the target; they are not predictive guarantees.

Followup	Budget	Rule	Classified	Errors	Requests	Selection regret (%)
MEAN	12	fixed depth	17/25	0	204	0.0000
MEAN	12	adaptive	9/25	0	204	0.0000
MEAN	18	calibration	6/25	6	240	0.0000
MEAN	18	immediate	25/25	0	42	0.0000
MEAN	18	fixed depth	25/25	0	252	0.0000
MEAN	18	adaptive	25/25	0	252	0.0000
EI	12	fixed depth	17/25	0	204	0.0000
EI	12	adaptive	9/25	0	204	0.0000
EI	18	calibration	6/25	6	240	0.2983
EI	18	immediate	25/25	1	42	0.0000
EI	18	fixed depth	25/25	0	252	0.0000
EI	18	adaptive	25/25	0	252	0.0000

Table S37. Small thermal model: audit gate versus simple selector and fixed policies at 96 calls per workflow. Paired difference is 100(audit/comparator-1); negative favours audit. This authored 24x24 finite-volume verification is not an industrial expense benchmark.

Source	Comparator	Median difference (%)	Win/tie/loss
central source	simple selector	0.000	0/9/1
central source	fixed mean	3.961	1/3/6
central source	fixed ei	4.961	2/2/6
central source	fixed lcb	4.961	2/2/6
offset source	simple selector	0.000	0/10/0
offset source	fixed mean	0.202	4/1/5
offset source	fixed ei	2.350	2/2/6
offset source	fixed lcb	2.350	1/3/6

Table S38. Selected fixed endpoint-design checks. All 14 h5 classes persist in six mesh/mapping contexts; 8 pairs have different endpoint designs and 6 are same-design ties. Selection was frozen from known contrasts and is not a new independent optimization sample. Fine means 200x100.

Case	Selection group	Original class	Native fine	Transferred fine	Endpoints
S01	changed	better	better	better	different
S02	changed	better	better	better	different
S03	changed	worse	worse	worse	different
S04	changed	worse	worse	worse	different
S05	changed	worse	worse	worse	different
S06	changed	worse	worse	worse	different
S07	changed	better	better	better	different
S08	independent proxy mismatch	worse	worse	worse	different
S09	unchanged different action	tie	tie	tie	same
S10	unchanged different action	tie	tie	tie	same
S11	unchanged different action	tie	tie	tie	same
S12	unchanged different action	tie	tie	tie	same
S13	structural alias	tie	tie	tie	same
S14	structural alias	tie	tie	tie	same

Table S39. Numerical flags and absolute objective variation for 104 selected design points. Recovery records retain the original failed gate. Objective differences are not known solver errors; fixed-design ranking stability does not imply physical convergence.

Check	Measured record
Original strict source-response gate	Failed on fourth paid attempt; threshold 1e-8 unchanged
Coarse source agreement flags	1/104 unique points
Flagged coarse relative discrepancy	6.7982718e-08
Maximum equilibrium relative residual	6.1852705e-07
Maximum energy/work relative gap	3.5447877e-07
Maximum native 200x100 objective difference	26.5739%
Maximum transferred 200x100 objective difference	32.0590%
Fine-grid optimization / convergence certification	Neither performed nor established

Table S40. Updated cumulative paid physics-attempt ledger. FE and thermal solves are both counted; recovery adds cost, not independent samples. Source/calibration GP times in the budget release were not separately recorded, so this is not an end-to-end speedup measurement.

Stage	Paid attempts	Boundary
Earlier GP/transfer stages	36480	Includes both 18120-call transfer executions
Archived common continuation	282	273 original attempts plus 9 record recovery attempts
Thermal verification	15888	7200 retained +22 checks +1080 recovery +7586 earlier unavailable
Independent budget verification	2134	All initialization/source/probe/oracle/mesh calls
Selected paired-grid verification	521	4 stopped-run attempts +517 recovery-execution attempts
Auxiliary physical-bound development	2	Reference solves only; no new optimizer samples
Full revision cumulative paid attempts	55307	Includes auxiliary development; earlier historical experiments excluded

Table S41. Complete seed-block composition. Each block has three dependent source states. K=1/2/3 counts states by exact branch-class count. The B=12 completion counts coincide for mean/EI continuation; blocks with zero contrasts remain in the record. No extra independent samples are introduced.

Block	Contrasts	K=1/2/3	Mean B/T/W	EI B/T/W	Depth B12	Adaptive B12
10	0	3/0/0	0/0/0	0/0/0	0	0
11	3	0/3/0	0/3/0	0/3/0	3	3
12	2	1/2/0	0/2/0	0/2/0	2	2
13	0	3/0/0	0/0/0	0/0/0	0	0
14	6	0/0/3	0/6/0	0/6/0	3	0
15	4	0/2/1	0/3/1	0/3/1	3	2
16	0	3/0/0	0/0/0	0/0/0	0	0
17	5	0/1/2	0/5/0	0/5/0	3	1
18	5	0/1/2	0/5/0	0/4/1	3	1
19	0	3/0/0	0/0/0	0/0/0	0	0

Table S42. Post-review descriptive sensitivity at all previously specified budgets and margins. Fixed-depth-reversed changes only the nonbaseline class order. All displayed cells have zero measured-target errors and zero reported state-selection loss; these are not predictive guarantees. New points sum statewise points absent from that state's training archive, not globally unique paid solves. Original fixed-depth/adaptive cells match the published values.

Followup	B	Margin	Order / rule	Classified	Requests	New points	Worse abstained
EI	12	0.0001	adaptive	9/25	204	152	1
EI	12	0.0001	fixed depth	17/25	204	156	0
EI	12	0.0001	fixed depth reversed	17/25	204	150	1
EI	12	0.001	adaptive	9/25	204	152	1
EI	12	0.001	fixed depth	17/25	204	156	0
EI	12	0.001	fixed depth reversed	17/25	204	150	1
EI	12	0.01	adaptive	9/25	204	152	1
EI	12	0.01	fixed depth	17/25	204	156	0
EI	12	0.01	fixed depth reversed	17/25	204	150	1
EI	18	0.0001	adaptive	25/25	252	174	0
EI	18	0.0001	fixed depth	25/25	252	174	0
EI	18	0.0001	fixed depth reversed	25/25	252	174	0
EI	18	0.001	adaptive	25/25	252	174	0
EI	18	0.001	fixed depth	25/25	252	174	0
EI	18	0.001	fixed depth reversed	25/25	252	174	0
EI	18	0.01	adaptive	25/25	252	174	0
EI	18	0.01	fixed depth	25/25	252	174	0
EI	18	0.01	fixed depth reversed	25/25	252	174	0
EI	24	0.0001	adaptive	25/25	252	174	0
EI	24	0.0001	fixed depth	25/25	252	174	0
EI	24	0.0001	fixed depth reversed	25/25	252	174	0
EI	24	0.001	adaptive	25/25	252	174	0
EI	24	0.001	fixed depth	25/25	252	174	0
EI	24	0.001	fixed depth reversed	25/25	252	174	0
EI	24	0.01	adaptive	25/25	252	174	0
EI	24	0.01	fixed depth	25/25	252	174	0
EI	24	0.01	fixed depth reversed	25/25	252	174	0
MEAN	12	0.0001	adaptive	9/25	204	144	0
MEAN	12	0.0001	fixed depth	17/25	204	141	0
MEAN	12	0.0001	fixed depth reversed	17/25	204	137	0
MEAN	12	0.001	adaptive	9/25	204	144	0
MEAN	12	0.001	fixed depth	17/25	204	141	0
MEAN	12	0.001	fixed depth reversed	17/25	204	137	0
MEAN	12	0.01	adaptive	9/25	204	144	0
MEAN	12	0.01	fixed depth	17/25	204	141	0
MEAN	12	0.01	fixed depth reversed	17/25	204	137	0

Followup	B	Margin	Order / rule	Classified	Requests	New points	Worse abstained
MEAN	18	0.0001	adaptive	25/25	252	152	0
MEAN	18	0.0001	fixed depth	25/25	252	152	0
MEAN	18	0.0001	fixed depth reversed	25/25	252	152	0
MEAN	18	0.001	adaptive	25/25	252	152	0
MEAN	18	0.001	fixed depth	25/25	252	152	0
MEAN	18	0.001	fixed depth reversed	25/25	252	152	0
MEAN	18	0.01	adaptive	25/25	252	152	0
MEAN	18	0.01	fixed depth	25/25	252	152	0
MEAN	18	0.01	fixed depth reversed	25/25	252	152	0
MEAN	24	0.0001	adaptive	25/25	252	152	0
MEAN	24	0.0001	fixed depth	25/25	252	152	0
MEAN	24	0.0001	fixed depth reversed	25/25	252	152	0
MEAN	24	0.001	adaptive	25/25	252	152	0
MEAN	24	0.001	fixed depth	25/25	252	152	0
MEAN	24	0.001	fixed depth reversed	25/25	252	152	0
MEAN	24	0.01	adaptive	25/25	252	152	0
MEAN	24	0.01	fixed depth	25/25	252	152	0
MEAN	24	0.01	fixed depth reversed	25/25	252	152	0

Table S43. B=12, margin 0.001: exhaustively permuting the alternatives within each state preserves completion coverage but can change which target remains unmeasured. Both continuations change the covered class set in eight states; none contains a better primary target. This is retrospective replay, not independent confirmation. The capacity of 17 completed nonbaseline comparisons follows from the declared six-request full-branch contract.

Followup	States changed	Classified	Worse missed range	Forward new points	Reverse new points	Max loss (%)
MEAN	8	17/25	0–0	141	137	0.0000
EI	8	17/25	0–1	156	150	0.0000

Table S44. Ranges of h5 baseline-minus-alternative paired differences for the eight differing endpoint pairs across six mesh/mapping contexts. Percentages are normalized by each context's initial incumbent compliance, not its baseline final endpoint. Six identical-endpoint ties are omitted here and retained in the original case table. These selected records neither add independent optimization samples nor certify a physical error bound.

Case	Existing state	Control	Class	Minimum (%)	Maximum (%)
S01	ei_r1_t101	EI	better	53.6937	54.4324
S02	ei_r1_t101	PI	better	12.4450	13.5994
S03	ei_r4_t101	PI	worse	-8.5397	-7.0706
S04	ucb_r7_t101	EI	worse	-13.6116	-11.8465
S05	ucb_r7_t101	PI	worse	-15.6248	-14.4497
S06	ucb_r7_t101	LCB	worse	-26.5189	-25.8253
S07	ucb_r8_t101	LCB	better	20.9040	23.8275
S08	beam_holdout_r18_pi	EI	worse	-10.5882	-8.3300

Table S45. Retrospective operational comparators answering the same conditional h5 question at margin 0.001. Named and exact-class full-horizon readers are transparent baselines, not published-algorithm replications. The strongest simple class-aware fixed-depth comparator is the frozen v1 method itself. Alias-skipping calibration is post hoc and leaves all labels unchanged; original results are retained. Requests include cache reads and are not new physical attempts. No predictive or deployment advantage is inferred. Data table assembled using OpenAI Codex (GPT-6)-assisted Python.

Followup	Reader / rule	Classified	Errors	Requests	Selection loss (%)
MEAN	named full horizon	25/25	0	720	0.0000
MEAN	exact class full horizon	25/25	0	330	0.0000
MEAN	fixed depth v1 B18	25/25	0	252	0.0000
MEAN	immediate v1 B18	25/25	0	42	0.0000
MEAN	adaptive v1 B18	25/25	0	252	0.0000
MEAN	calibration original	6/25	6	240	0.0000
MEAN	calibration skip alias posthoc	6/25	6	136	0.0000
EI	named full horizon	25/25	0	720	0.0000
EI	exact class full horizon	25/25	0	330	0.0000
EI	fixed depth v1 B18	25/25	0	252	0.0000
EI	immediate v1 B18	25/25	1	42	0.0000
EI	adaptive v1 B18	25/25	0	252	0.0000
EI	calibration original	6/25	6	240	0.2983
EI	calibration skip alias posthoc	6/25	6	136	0.2983

S11. Checked opportunity, development costs and assumption boundaries

The additional checks read losslessly restored frozen records and existing derived summaries. They retain 60 metadata states, 90 dependent contexts and ten original seed groups per dataset. All 179 archived file hashes match. No new physical evaluation, bound query, GP fit, optimization sample or confirmatory test family is acquired. These are post hoc development and descriptive checks, not new independent validation.

S11.1. *Opportunity and branch-shape scope*

At its original 10^{-6} margin, archived common continuation has 3 better, 34 tie and 9 worse distinct nonbaseline contrasts, with better branches in only two states. Both independent continuations retain their 0.001 margin and have zero better branches; all statewise best-observed improvements are exactly zero. Complete horizon measurement therefore cannot demonstrate positive-case prediction or avoidance of false rejection.

Observed gains need not have diminishing increments. The check below includes the initial-to-h0 increment and the following five increments, allowing relative numerical tolerance 10^{-12}. It concerns deterministic recorded paths, not stochastic expected loss, and is not a replication or rejection of published model-based bandit algorithms.

Table S46. Descriptive shape check of distinct recorded branches. Each row retains 30 states and ten complete groups.

Record	Curves	Violating curves	States with a violation
Archived mean	76	9	6
Previous mean	55	10	5
Previous ei	55	3	2

S11.2. *Audit, certification and chosen-branch execution*

Table S47 separates audit requests, chosen nonalias completion and alias-only execution. Previously quoted nontrivial-state totals 252, 204 and 402 become 330, 282 and 426 when all original states are executed. These totals do not mean every unobserved competitor is completed or every selection is certified. Selection certification coverage is shown separately. They are logical request costs, including cache reads, not new physical solves or end-to-end wall-time savings.

Table S47. Original monotone development costs and certified-selection coverage, retaining all 30 states per row. Coverage in the last column excludes alias-only states.

Record / budget / reader	Classified	Audit	Chosen nonalias completion	Alias execution	Total	Certified selection
Archived mean / B18 / fixed	41/46	402	0	24	426	21/26
Archived mean / B18 / h0-directed	36/46	402	0	24	426	21/26
Archived mean / B18 / alternative first	31/46	393	39	24	456	25/26
Previous mean / B18 / fixed	25/25	252	0	78	330	17/17
Previous mean / B18 / h0-directed	25/25	247	5	78	330	17/17
Previous mean / B18 / alternative first	25/25	247	5	78	330	17/17
Previous mean / B12 / fixed	17/25	204	0	78	282	9/17
Previous mean / B12 / h0-directed	9/25	199	5	78	282	9/17
Previous ei / B18 / fixed	25/25	252	0	78	330	17/17
Previous ei / B18 / h0-directed	25/25	247	5	78	330	17/17
Previous ei / B18 / alternative first	25/25	247	5	78	330	17/17
Previous ei / B12 / fixed	17/25	204	0	78	282	9/17
Previous ei / B12 / h0-directed	9/25	199	5	78	282	9/17

Under independent terminal intervals $[L_j, U_j]$, the worst-case chosen-branch loss is $\max(0, U_i - \min_{j \neq i} L_j)$. With an unfinished competitor lower bound zero, a chosen upper bound above τm_{init} cannot certify the loss tolerance. Every recorded branch terminal incumbent exceeds that threshold, so all its earlier prefixes do as well. Within this interval-only information model, certifying every selection and then executing the chosen branch returns to six requests per exact class: 456 over archived states and 330 per previous continuation. This is a scope-limited accounting barrier, not a new theorem for all physical or GP-informed audits. Structural alias choices have zero selection regret without an audit, but executing their branch still costs six requests.

The completed two-reference/full-pool screen separately fails necessary root-tie sharpness on all 43 nontrivial states. Its 1,720 bound queries and two reference solves remain original development costs. A sampled minimum can disqualify the full envelope but cannot certify it. Fixed-kernel GP reachability has a shorter H1 specification; no H1 result is used here as H5 efficacy or independent validation.

The checks and source hashes are supplied with `innovation_readiness_20261014/`.

The reported experiment/reference ledger contains 55,307 physics attempts within its stated scope. This descriptive preparation adds zero. No physical accuracy of the illustrative margin, independent positive-case efficacy or journal novelty sufficiency is certified.

S12. GP proof verification and exact envelope completion

The attached immutable GP execution archive has SHA-256 `25501c61eedfb379312693829be3fe612c2285ed5f2e7e677c3dc4892ca8b139`. Its 70 manifest-listed members, 204 frozen source files, 43 construction records and two completed witnesses are verified. All60 metadata states and complete ten-group datasets remain retained. The two attempts are consumed; there is no retry, replacement, recovery or new optimization sample in the present verification.

S12.1. *Scope of the affine and numerical certificates*

The affine-mean equation in the main text describes the exact solution of the supplied binary kernel matrix. For an approximate inverse $\widehat{P}$, a verified norm residual below one implies invertibility through the Neumann-series bound. If $\widehat{w}_i$ has residual $r_i = k_i - A\widehat{w}_i$, its error is bounded by $\|A^{-1}\|_\infty \|r_i\|_\infty$. The normalization correction contributes at most the same maximum weight error; the effective coefficients use twice that bound plus enclosed arithmetic. The serialized coefficient bands are endpoints of binary numbers. All114 reconstructed matrices and cross matrices have the original byte hashes; reapplying the original enclosures reproduces every stored coefficient and error array exactly.

The new pruner solves the necessary inequalities exactly over $[\ell, \infty)$. It reads no pending response and introduces no label. This is post hoc algorithm development on previously constructed bands, not a prospectively independent study. It does not certify the kernel's transcendental evaluation error or every native floating GP prediction for every unknown label. The mathematical interval may be an outer relaxation of the coupled coefficient uncertainty.

S12.2. *Why the certificate has at most two rival indices*

Each necessary pair inequality is $(a_i^- - a_j^+)z \le b_j^+ - b_i^-$. It imposes an upper half-line, a lower half-line, no constraint, or an immediate parallel contradiction. An empty intersection with $z \ge \ell$ therefore has a contradiction from one upper/domain pair, one parallel pair, or one lower and one upper bound. The envelope identifies these constraints at its maximizing gap location. Its separate checker uses only the indicated binary coefficients and exact rational subtraction/division. It neither trusts a floating intersection nor reconstructs the hull.

The envelope is the lower envelope of *upper-band* lines. Every original lower-band line is queried afterwards. Removing a line from the envelope does not remove its candidate index. This distinction preserves an action that minimizes only at one response. For example, $\mu_0 = 1$, $\mu_1 = z$ and $\mu_2 = 2 - z$ give action0 the singleton interval $\{1\}$; it is the lowest-index argmin at that response. This is a synthetic unit example, not an added engineering optimization sample. Classical acceptance-set geometry is acknowledged in the main bibliography; no first-ever affine-envelope claim is made.

S12.3. *Paired construction times and complete cost scope*

The frozen pairwise reader is the same-contract reference, not a state-of-the-art library benchmark. Both algorithms receive identical coefficient centers and outward error bands. Three repeated timings per branch/domain alternate algorithm order; casewise medians are summed. They give 19.257763 versus 6.040865 seconds, ratio 3.187915. Complete independent proof replays separately require 3.578688 versus 2.668741 seconds. These single replay passes are additional component measurements; their combined ratio is 2.621984. Machine load, implementations, thread settings and versions limit transfer of the absolute times.

Shared GP coefficient work, original initial data and baseline endpoints are not charged as if avoided. The earlier execution paid44,146 unique state/design physical-bound computations and176.433648 seconds of construction. Its GP coefficient/outer-set component took21.019126 seconds and rational construction replay4.000773 seconds; grading took5.936882 seconds. The new benchmark/reconstruction is additional verification work, not a deduction from those historical elapsed times. No end-to-end rerun is performed and no realized deployment saving is claimed. Logical reader requests116,850 versus568, and unique cached requests44,146 versus498, remain counterfactual for the subset reader.

S12.4. *H1 and H5 interpretation*

Both domains include114/114 stored next actions. All232,564 original and232,564 new exclusions pass independent rational checking. The new exact intervals are subsets of the original outward intervals; regrading them leaves all71 nonbaseline H1 contrasts per domain abstaining and all43 baseline selections uncertified. The H1 completed baseline is already paid and fallback is not a selection certificate. H5 reachability is not implemented. The original independent-study gate remains false, while the computational-stage increment is separately verified.

Runtime for this check is Python3.12, NumPy2.3.5, SciPy1.17.0 and scikit-learn1.8.0 with one BLAS thread. New physical attempts, physical-bound queries, hyperparameter fits, independent optimization samples and confirmatory families are zero. Two previous witness attempts bring the reported project total to55,309. Stored execution and verification records identify exact source/output hashes and dependent reporting units.

S13. Current evidence availability and historical paired-grid limitation

The current contribution can be checked from the supplied GP execution archive and the new coefficient-band certificates without a finite-element solver. The execution archive retains both frozen witnesses and their original attempt accounting, all construction records and its file manifest. The separate envelope verifier reconstructs all 114 binary kernel systems, compares all 228 branch-domain sets and checks 232,564 new exclusions. The existing test record reports 31 scoped unit tests. The computational comparison is post hoc development, and its three repetitions and two response domains do not add independent optimization samples.

The supporting paired-grid archive has a different preservation status. Its repository provenance audit lists 25 exact JSON/JSONL files. Twenty-two files are currently recovered and match those recorded hashes. Three completed-recovery files are not recovered:

- `events.jsonl`: 566,891 bytes; SHA-256:
  `5c526b95919b6db0cd145bd78dc906f93457e10ac6bf8b0b99de7d45ec88c222.`
- `responses.json`: 341,628 bytes; SHA-256:
  `d8076558c9abf1ba6cd279320ed7fa2618bbe38617f12a2e0b0ad563d393e9e2.`
- `run_status.json`: 389 bytes; SHA-256:
  `beeb6072b9ea737617686064a64caf8e934f20cc8d4e45145644f4b3b305e65d.`

These files belong to `results/engineering_optimization_paired_mesh_recovery_v1/`. No copy is present in the checked baseline tree or scoped attachments, and the user reports the selected directory absent from the local backup checked. This is not proof that every other saved archive lacks a copy. The missing complete base85 archive had one manifest and six segments; its old full review-package manifest records their hashes.

The preserved `analysis.json`, `paired_contrasts.csv` and `case_summary.csv` match the historical full review-package member hashes. They permit review of the reported contrasts and aggregate summaries, but do not permit a complete new validation of solver residuals, ordered request/label events, density replays or cache provenance. Earlier sections describe the completed historical study, not a newly performed full raw-record replay. The original source-agreement failure, 521 paid attempts and missing original response value/duration remain disclosed. Recovered final records would contain the historical remeasurement, not restore that absent original value or timing.

The missing files are not reconstructed from summary tables and no solver is rerun to fill the gap. The archive limitation is kept separate from the complete current GP action-set verification. It narrows the reproducibility claim for supporting paired-grid observations without changing their recorded values or promoting them to physical-error or convergence certificates.

S14. H1 threshold decomposition and contribution closure

This descriptive analysis reads the existing exact action intervals, point bounds and paid baseline endpoints. It reads no additional physical response, evaluates no physical bound and changes no protocol. For branch threshold $T = B - \epsilon$, the main-text joint lower bound is at least T exactly when the initial incumbent and both terms for every retained action are at least T. The proof is the elementary equivalence between a minimum exceeding a threshold and all of its arguments exceeding that threshold. The result describes sufficiency of the supplied relaxed certificate; it is not an impossibility result for the coupled GP or true engineering objective.

Both predeclared domains give71 nonbaseline contrasts and281 retained action terms. All71 contrasts have at least one pending term and one next-point term below threshold. There are71 failing pending terms and276 failing next-point terms. These are dependent descriptive counts; the two domains do not double the independent sample size. The median weakest next-point-bound/threshold ratio is0.1622233735904155. Raising only pending terms, or only next-point terms, cannot make the fixed-set certificate pass. Actual stronger physical information could change response domains or action sets, so no extrapolated impossibility claim is made.

The reader is `src/diagnose_h1_bound_bottlenecks.py`. Its output includes branch- and action-level rational thresholds, input SHA-256 hashes and a summary. Four synthetic checks cover the closed margin boundary, separate information bottlenecks, distinct blocking actions and the independent incumbent constraint. These checks contain no engineering optimization samples. The original31-test record remains separate.

The contribution judgment remains limited: numerical certificate production and its scoped computational improvement are supported; positive independent decision efficacy and full deployment savings are not. This additional diagnosis locates the physical-information bottleneck, but does not turn that negative outcome into an optimizer-performance claim or certify a journal novelty threshold.

References

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